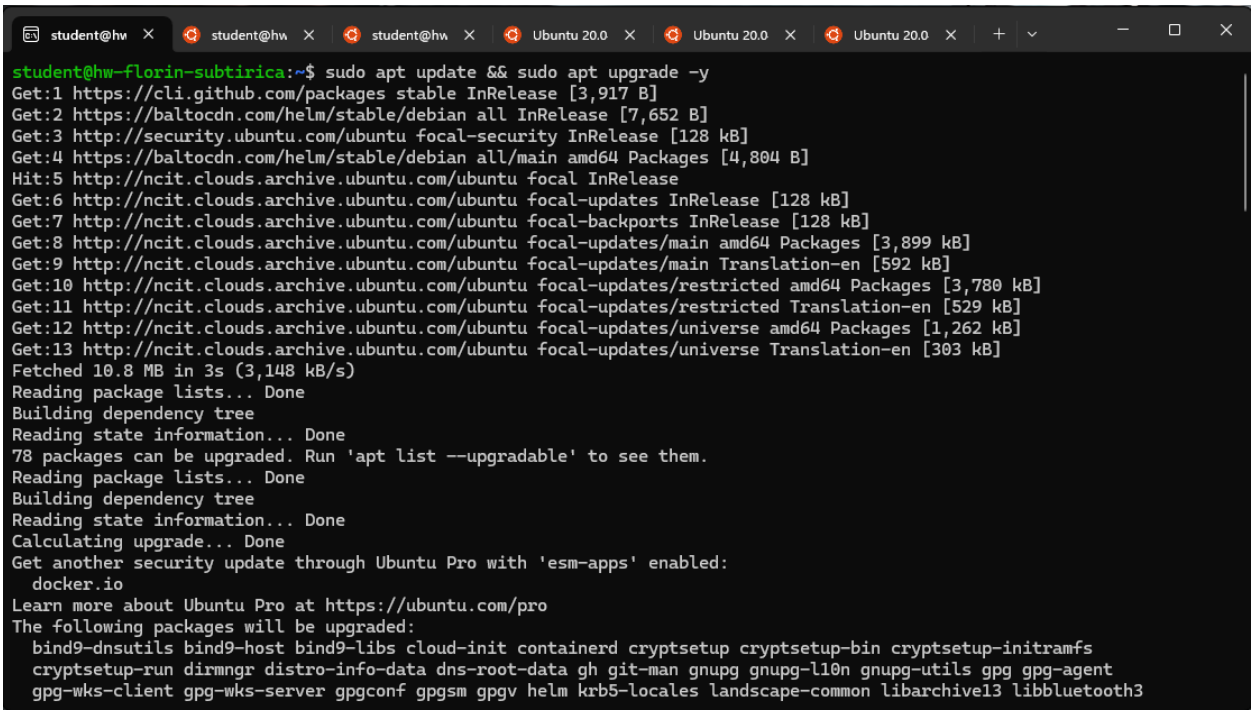
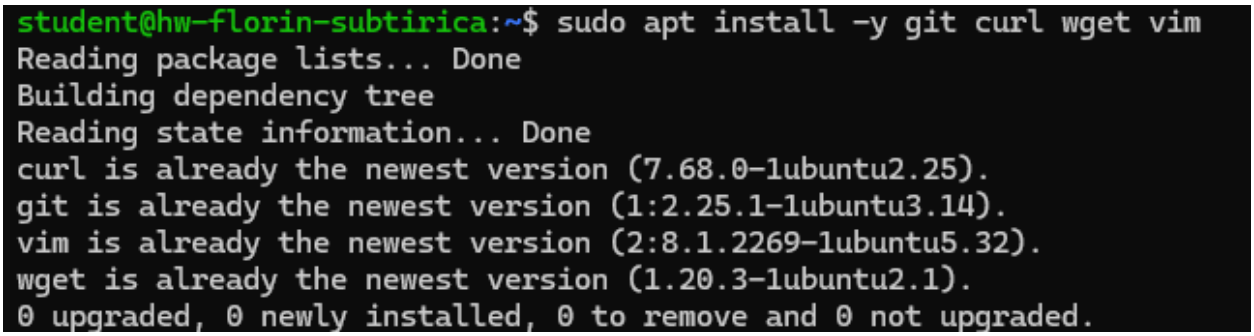
Task 0:



Am dat update și upgrade.



Am instalat curl, gitm wget, vim (erau deja instalate).

```
student@hw-florin-subtirica:~$ docker

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build       Build an image from a Dockerfile
  pull        Download an image from a registry
  push        Upload an image to a registry
```

Am instalat docker (era deja instalat).

```
student@hw-florin-subtirica:~$ kubectl
kubectl controls the Kubernetes cluster manager.

 Find more information at: https://kubernetes.io/docs/reference/kubectl/

Basic Commands (Beginner):
  create          Create a resource from a file or from stdin
  expose          Take a replication controller, service, deployment or pod and expose it as a new Kubernetes service
  run             Run a particular image on the cluster
  set             Set specific features on objects

Basic Commands (Intermediate):
  explain         Get documentation for a resource
  get             Display one or many resources
  edit            Edit a resource on the server
```

Am instalat kubectl (era deja instalat).

```
student@hw-florin-subtirica:~$ kind

Command 'kind' not found, did you mean:

  command 'kid' from deb python-kid (0.9.6-3build1)
  command 'find' from deb findutils (4.7.0-1ubuntu1)
  command 'king' from deb king (2.23.161103+dfsg1-3)
  command 'kino' from deb kino (1.3.4+dfsg0-1build1)
  command 'kfind' from deb kfind (4:19.12.3-0ubuntu1)

Try: sudo apt install <deb name>

student@hw-florin-subtirica:~$ curl -Lo ./kind https://kind.sigs.k8s.io/dl/latest/kind-linux-amd64
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100    86  100    86    0     0    289      0 --:--:-- --:--:-- --:--:--   289
100 10.5M  100 10.5M    0     0  5351k      0  0:00:02  0:00:02 --:--:-- 8419k
student@hw-florin-subtirica:~$ chmod +x ./kind
student@hw-florin-subtirica:~$ sudo mv ./kind /usr/local/bin/
student@hw-florin-subtirica:~$ kind
kind creates and manages local Kubernetes clusters using Docker container 'nodes'

Usage:
  kind [command]

Available Commands:
  build       Build one of [node-image]
  completion  Output shell completion code for the specified shell (bash, zsh or fish)
  create      Creates one of [cluster]
```

Am instalat kind.

Task 1:

```
student@hw-florin-subtirica:~$ kind create cluster --name hw-cluster
Creating cluster "hw-cluster" ...
 ✓ Ensuring node image (kindest/node:v1.33.0) 🖼
 ✓ Preparing nodes 📦
 ✓ Writing configuration 📜
 ✓ Starting control-plane 🕹️
 ✓ Installing CNI 🔌
 ✓ Installing StorageClass 💾
Set kubectl context to "kind-hw-cluster"
You can now use your cluster with:

kubectl cluster-info --context kind-hw-cluster

Have a question, bug, or feature request? Let us know! https://kind.sigs.k8s.io/#community 🙂
student@hw-florin-subtirica:~$
```

Am creat cluster-ul hw-cluster folosind kind.

```
student@hw-florin-subtirica:~$ kubectl config current-context
kind-hw-cluster
```

Am verificat existența acestuia.

```
student@hw-florin-subtirica:~$ kubectl get nodes
NAME                       STATUS   ROLES           AGE   VERSION
hw-cluster-control-plane   Ready    control-plane   95s   v1.33.0
student@hw-florin-subtirica:~$ kubectl get pods -n kube-system
NAME                                               READY   STATUS    RESTARTS   AGE
coredns-674b8bbfcf-qmwnw                           1/1     Running   0          90s
coredns-674b8bbfcf-vc47p                           1/1     Running   0          90s
etcd-hw-cluster-control-plane                      1/1     Running   0          97s
kindnet-6w844                                      1/1     Running   0          90s
kube-apiserver-hw-cluster-control-plane            1/1     Running   0          97s
kube-controller-manager-hw-cluster-control-plane   1/1     Running   0          97s
kube-proxy-5vcz2                                   1/1     Running   0          90s
kube-scheduler-hw-cluster-control-plane            1/1     Running   0          97s
student@hw-florin-subtirica:~$
```

Task 2:

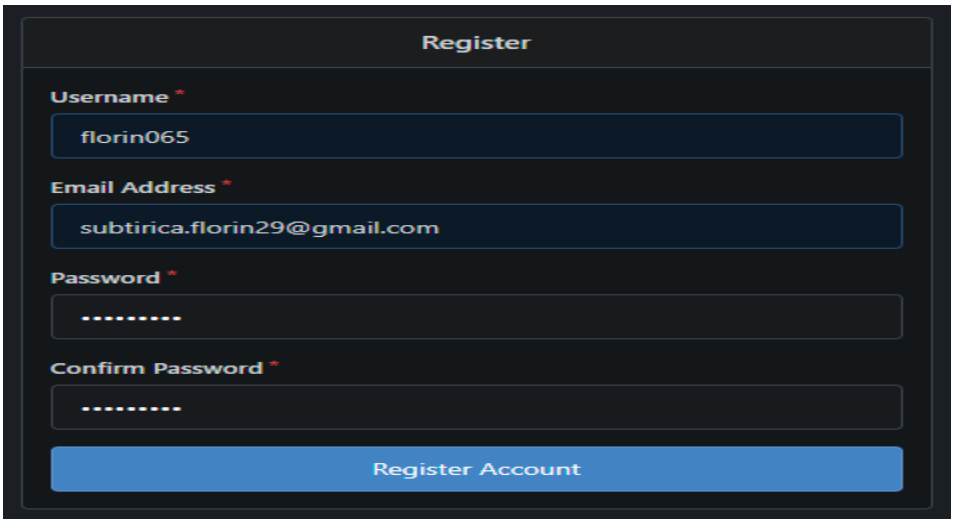
```
student@hw-florin-subtirica:~$ kubectl create namespace gitea
namespace/gitea created
student@hw-florin-subtirica:~$ helm repo add gitea-charts https://dl.gitea.io/charts/
 update
"gitea-charts" has been added to your repositories
student@hw-florin-subtirica:~$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "gitea-charts" chart repository
Update Complete. ⎈Happy Helming!⎈
student@hw-florin-subtirica:~$ helm install gitea gitea-charts/gitea --namespace gitea
NAME: gitea
LAST DEPLOYED: Tue Apr 29 15:36:11 2025
NAMESPACE: gitea
STATUS: deployed
REVISION: 1
NOTES:
1. Get the application URL by running these commands:
  echo "Visit http://127.0.0.1:3000 to use your application"
  kubectl --namespace gitea port-forward svc/gitea-http 3000:3000
student@hw-florin-subtirica:~$ kubectl --namespace gitea port-forward svc/gitea-http 3000:3000
error: unable to forward port because pod is not running. Current status=Pending
student@hw-florin-subtirica:~$ kubectl --namespace gitea port-forward svc/gitea-http 3000:3000
error: unable to forward port because pod is not running. Current status=Pending
student@hw-florin-subtirica:~$ kubectl --namespace gitea port-forward svc/gitea-http 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
```

Am creat namespace-ul pentru gitea, am instalat gitea folosind helm și am făcut port forward.

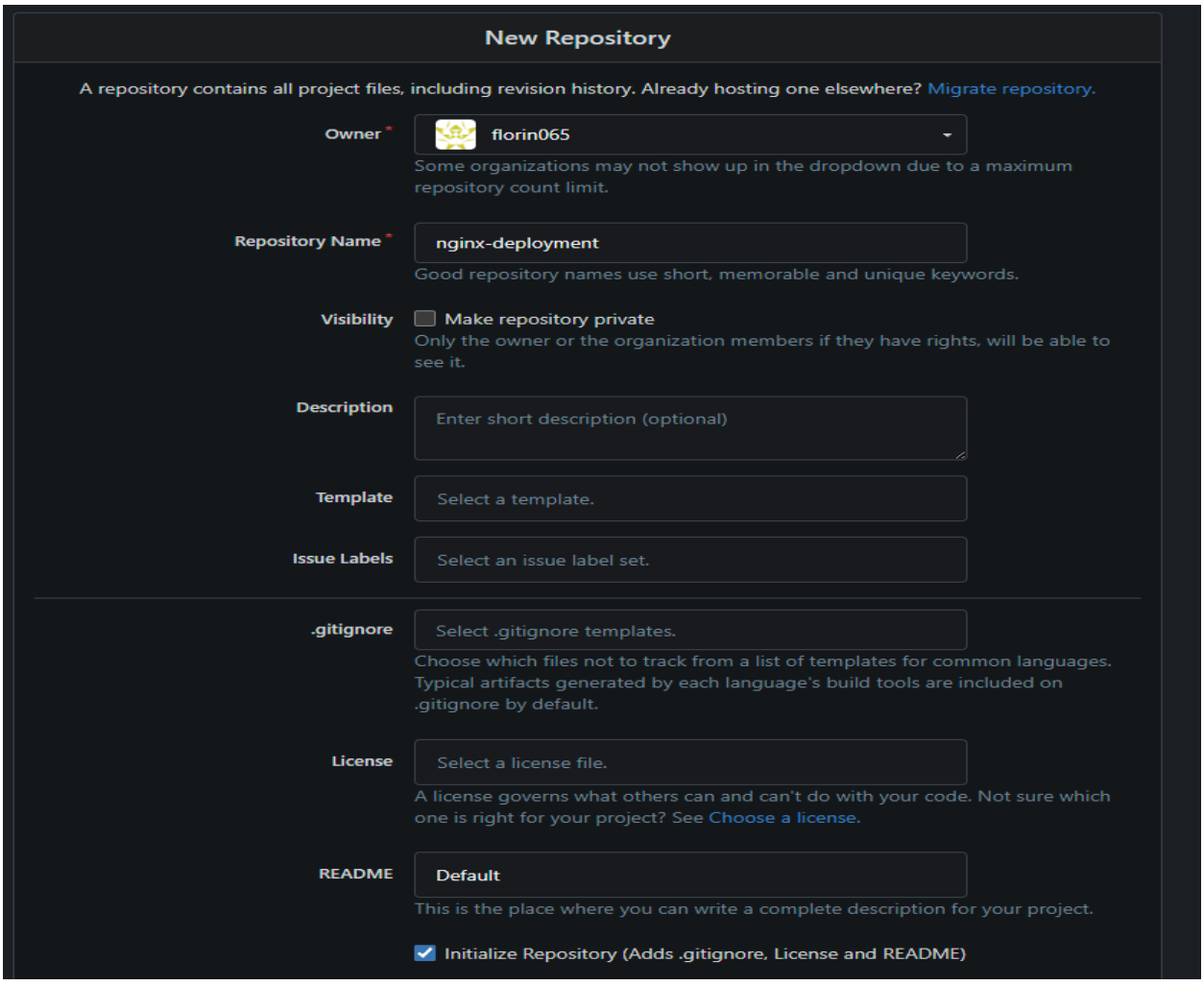
```
florin@FLORIN ~   X ssh -J fep -L 3000:127.0.0.1:3000 -i ~/.ssh/id_rsa  student@10.9.3.28

* IMPORTANT - PLEASE READ CAREFULLY:
* Începând cu 28.09.2023 vom muta directoarele home vechi la calea $OLD_HOME. Vă rugăm să mutați
ce mai este relevant în home-ul cel nou pana pe 05.10.2023
* If you are using GPU computing and TensorFlow technology, please use CUDA_VISIBLE_DEVICES
  to limit the number of GPUs. Otherwise, you will be banned.
* Your sessions will be disconnected automatically after 60min on inactivity.
  You can use "-o ServerAliveInterval 10" option if you need longer sessions.*
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-200-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
```



Am făcut cont pe gitea.



Am creat repo-ul numit nginx-deployment.

Task 3:

```
student@hw-florin-subtirica:~$ kubectl create namespace argocd
namespace/argocd created
student@hw-florin-subtirica:~$ kubectl apply -n argocd -f https://raw.githubusercontent.com/argoproj/argo-cd
/stable/manifests/install.yaml
customresourcedefinition.apiextensions.k8s.io/applications.argoproj.io created
customresourcedefinition.apiextensions.k8s.io/applicationsets.argoproj.io created
customresourcedefinition.apiextensions.k8s.io/appprojects.argoproj.io created
serviceaccount/argocd-application-controller created
serviceaccount/argocd-applicationset-controller created
serviceaccount/argocd-dex-server created
serviceaccount/argocd-notifications-controller created
serviceaccount/argocd-redis created
serviceaccount/argocd-repo-server created
serviceaccount/argocd-server created
role.rbac.authorization.k8s.io/argocd-application-controller created
role.rbac.authorization.k8s.io/argocd-applicationset-controller created
role.rbac.authorization.k8s.io/argocd-dex-server created
```

Am creat namespace-ul argocd și am instalat argocd.

```
student@hw-florin-subtirica:~$ kubectl port-forward svc/argocd-server -n argocd 8080:443
Forwarding from 127.0.0.1:8080 -> 8080
Forwarding from [::1]:8080 -> 8080
```

Am făcut port-forward.

```
florin@FLORIN ~  X ssh -J fep -L 8080:127.0.0.1:8080 -i ~/.ssh/id_rsa  student@10.9.3.28

* IMPORTANT - PLEASE READ CAREFULLY:
* Începând cu 28.09.2023 vom muta directoarele home vechi la calea $OLD_HOME. Vă rugăm să mutați
ce mai este relevant în home-ul cel nou pana pe 05.10.2023
* If you are using GPU computing and TensorFlow technology, please use CUDA_VISIBLE_DEVICES
  to limit the number of GPUs. Otherwise, you will be banned.
* Your sessions will be disconnected automatically after 60min on inactivity.
```

```
student@hw-florin-subtirica:~$ kubectl -n argocd get secret argocd-initial-admin-secret -o jsonpath="{.data.
password}" | base64 -d
MVqO4IHlL7Czcw56student@hw-florin-subtirica:~$ curl -sSL -o argocd https://github.com/argoproj/argo-cd/relea
ses/latest/download/argocd-linux-amd64
ocd
sudo mv argocd /usr/local/bin/
student@hw-florin-subtirica:~$ chmod +x argocd
student@hw-florin-subtirica:~$ sudo mv argocd /usr/local/bin/
student@hw-florin-subtirica:~$ argocd
argocd controls a Argo CD server

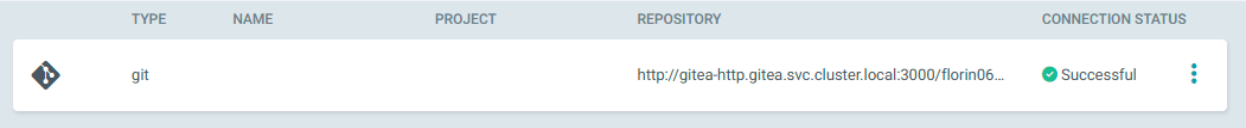
Usage:
  argocd [flags]
  argocd [command]

Available Commands:
  account     Manage account settings
  admin       Contains a set of commands useful for Argo CD administrators and requires direct Kubernetes ac
cess
  app         Manage applications
  appset      Manage ApplicationSets
  cert        Manage repository certificates and SSH known hosts entries
  cluster     Manage cluster credentials
```

Am aflat parola pentru argocd și am instalat argocd cli în terminal pentru a mă loga și a adăuga repo-ul de gitea. (din ui nu mergea)

```
student@hw-florin-subtirica:~$ argocd login localhost:8080 --username admin --password 'MVqO4IHlL7Czcw56'
WARNING: server certificate had error: tls: failed to verify certificate: x509: certificate signed by unknow
n authority. Proceed insecurely (y/n)? y
'admin:login' logged in successfully
Context 'localhost:8080' updated
```

```
student@hw-florin-subtirica:~$ argocd login localhost:8080 --username admin --password 'MVqO4IHlL7Czcw56'
WARNING: server certificate had error: tls: failed to verify certificate: x509: certificate signed by unknow
n authority. Proceed insecurely (y/n)? y
'admin:login' logged in successfully
Context 'localhost:8080' updated
student@hw-florin-subtirica:~$ argocd repo add http://gitea-http.gitea.svc.cluster.local:3000/florin065/ngin
x-deployment.git --username florin065 --password florin123
Repository 'http://gitea-http.gitea.svc.cluster.local:3000/florin065/nginx-deployment.git' added
```



Aici se poate vedea cum m-am logat și am adăugat repo-ul de gitea pe argocd.

Task 4:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: cloud-courses-deployment
  labels:
    app: cloud-courses
spec:
  replicas: 1
  selector:
    matchLabels:
      app: cloud-courses
  template:
    metadata:
      labels:
        app: cloud-courses
    spec:
      containers:
      - name: cloud-courses
        image: cloud-courses:latest
        imagePullPolicy: IfNotPresent
        ports:
        - containerPort: 3000
        env:
        - name: URL
          value: "http://localhost"
        - name: BASE_URL
          value: "/cloud-courses/"
---
apiVersion: v1
kind: Service
metadata:
  name: cloud-courses-service
spec:
  selector:
    app: cloud-courses
  ports:
  - protocol: TCP
    port: 8081
    targetPort: 3000
  type: NodePort
```

Manifest-ul pentru cloud courses.

```
student@hw-florin-subtirica:~/nginx-deployment$ vi deployment.yaml
student@hw-florin-subtirica:~/nginx-deployment$ git add deployment.yaml
student@hw-florin-subtirica:~/nginx-deployment$ git commit -m "add manifest to gitea"
[main 7fd4b9b] add manifest to gitea
 Committer: student <student@hw-florin-subtirica.grid.pub.ro>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 41 insertions(+)
 create mode 100644 deployment.yaml
student@hw-florin-subtirica:~/nginx-deployment$ git push
Username for 'http://localhost:3000': florin065
Password for 'http://florin065@localhost:3000':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 609 bytes | 609.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0)
remote: . Processing 1 references
remote: Processed 1 references in total
To http://localhost:3000/florin065/nginx-deployment.git
   5b7d420..7fd4b9b  main -> main
```

Aici i-am dat push pe gitea.

```yaml
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: cloud-courses-app
  namespace: argocd
spec:
  project: default
  source:
    repoURL: 'http://gitea-http.gitea.svc.cluster.local:3000/florin065/nginx-deployment.git'
    targetRevision: HEAD
    path: .
  destination:
    server: 'https://kubernetes.default.svc'
    namespace: default
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
    syncOptions:
    - CreateNamespace=true
```

```
student@hw-florin-subtirica:~/nginx-deployment$ vi app.yaml
student@hw-florin-subtirica:~/nginx-deployment$ kubectl apply -f app.yaml
application.argoproj.io/cloud-courses-app created
```

Aici este app.yaml și se vede că am dat apply.

```
student@hw-florin-subtirica:~/nginx-deployment$ git add .
student@hw-florin-subtirica:~/nginx-deployment$ git commit -m "add app and cloud courses to gitea"
[main 059a44e] add app and cloud courses to gitea
 Committer: student <student@hw-florin-subtirica.grid.pub.ro>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 289 files changed, 24988 insertions(+)
 create mode 100644 app.yaml
 create mode 100644 cloud-courses/.gitlab-ci.yml
 create mode 100644 cloud-courses/Dockerfile
 create mode 100644 cloud-courses/README.md
 create mode 100644 cloud-courses/docker-compose.yml
 create mode 100644 cloud-courses/website/.gitignore
 create mode 100644 cloud-courses/website/babel.config.js
 create mode 100644 cloud-courses/website/blog/2024-08-12-SOCcare.md
```

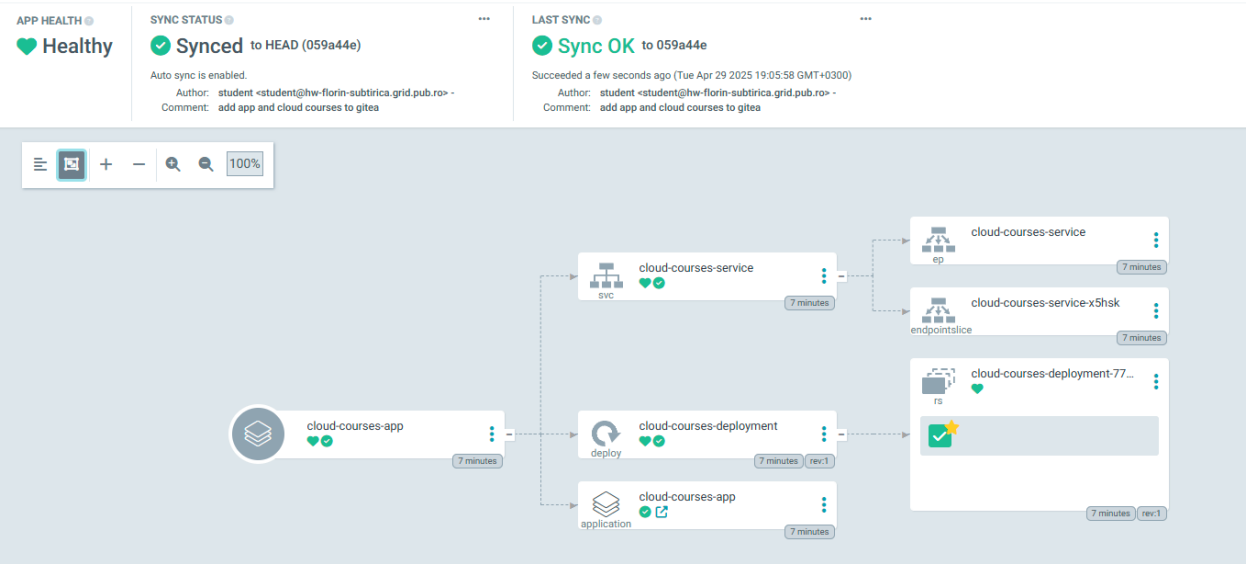
```
student@hw-florin-subtirica:~/nginx-deployment$ git push
Username for 'http://localhost:3000': florin065
Password for 'http://florin065@localhost:3000':
Enumerating objects: 346, done.
Counting objects: 100% (346/346), done.
Delta compression using up to 4 threads
Compressing objects: 100% (339/339), done.
Writing objects: 100% (345/345), 6.12 MiB | 15.66 MiB/s, done.
Total 345 (delta 29), reused 0 (delta 0)
remote: Resolving deltas: 100% (29/29), done.
remote: . Processing 1 references
remote: Processed 1 references in total
To http://localhost:3000/florin065/nginx-deployment.git
   7fd4b9b..059a44e  main -> main
```

```
student@hw-florin-subtirica:~/nginx-deployment$ cd cloud-courses/
student@hw-florin-subtirica:~/nginx-deployment/cloud-courses$ docker build -t cloud-courses:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  8.426MB
Step 1/8 : FROM node:lts-alpine
lts-alpine: Pulling from library/node
f18232174bc9: Pull complete
63786652eaff: Pull complete
0c2b42acd277: Pull complete
4509b69886a5: Pull complete
Digest: sha256:ad1aedbcc1b0575074a91ac146d6956476c1f9985994810e4ee02efd932a68fd
Status: Downloaded newer image for node:lts-alpine
 ---> 461edc13e56b
Step 2/8 : WORKDIR /app/website
 ---> Running in e9d130202182
 ---> Removed intermediate container e9d130202182
 ---> ef230bfc1044
Step 3/8 : EXPOSE 3000
```

```
student@hw-florin-subtirica:~/nginx-deployment/cloud-courses$ kind load docker-image cloud-courses:latest --name hw-cluster
Image: "cloud-courses:latest" with ID "sha256:34c53b440ca85bc73889674ffa21f335f686514e3e0ff079fd434758dd8dc36e" not yet present on node "hw-cluster-control-plane", loading...
```

Aici am dat build la imaginea de cloud courses și după care am dat kind load.

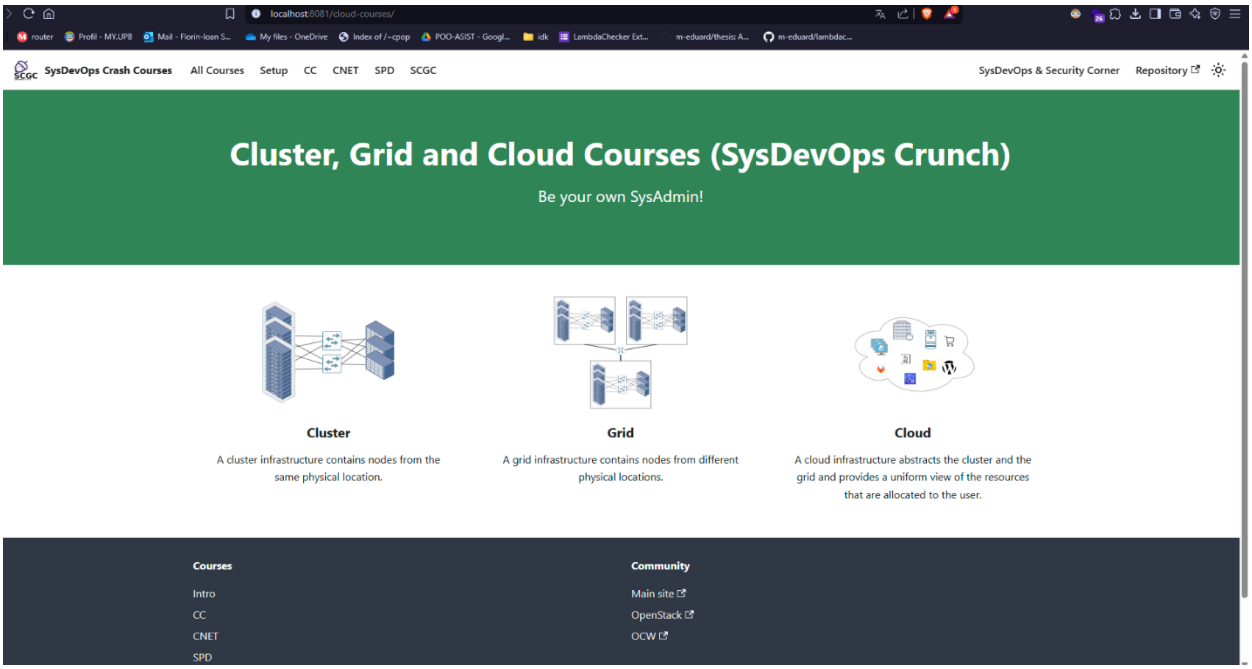


Pe argocd se poate vedea că totul a funcționat.

```
student@hw-florin-subtirica:~/nginx-deployment$ kubectl port-forward svc/cloud-courses-service 8081:8081
Forwarding from 127.0.0.1:8081 -> 3000
Forwarding from [::1]:8081 -> 3000
```

```
florin@FLORIN ~  X ssh -J fep -L 8081:127.0.0.1:8081 -i ~/.ssh/id_rsa  student@10.9.3.28

* IMPORTANT - PLEASE READ CAREFULLY:
* Începând cu 28.09.2023 vom muta directoarele home vechi la calea $OLD_HOME. Vă rugăm să mutați
ce mai este relevant în home-ul cel nou pana pe 05.10.2023
* If you are using GPU computing and TensorFlow technology, please use CUDA_VISIBLE_DEVICES
  to limit the number of GPUs. Otherwise, you will be banned.
* Your sessions will be disconnected automatically after 60min on inactivity.
  You can use "-o ServerAliveInterval 10" option if you need longer sessions.*
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-200-generic x86_64)
```



După care am făcut port forward pentru a accesa website-ul.

Task 5:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: cloud-courses-deployment
  labels:
    app: cloud-courses
spec:
  replicas: 1
  selector:
    matchLabels:
      app: cloud-courses
  template:
    metadata:
      labels:
        app: cloud-courses
    spec:
      containers:
      - name: cloud-courses
        image: cloud-courses:latest
        imagePullPolicy: IfNotPresent
        ports:
        - containerPort: 3000
        env:
        - name: URL
          value: "http://localhost"
        - name: BASE_URL
          value: "/cloud-courses/"
        livenessProbe:
          httpGet:
            path: "/cloud-courses/"
            port: 3000
          initialDelaySeconds: 5
          periodSeconds: 10
---
apiVersion: v1
kind: Service
metadata:
  name: cloud-courses-service
spec:
  selector:
    app: cloud-courses
  ports:
  - protocol: TCP
    port: 8081
    targetPort: 3000
  type: NodePort
```

Manifest-ul modificat pentru a avea liveness.

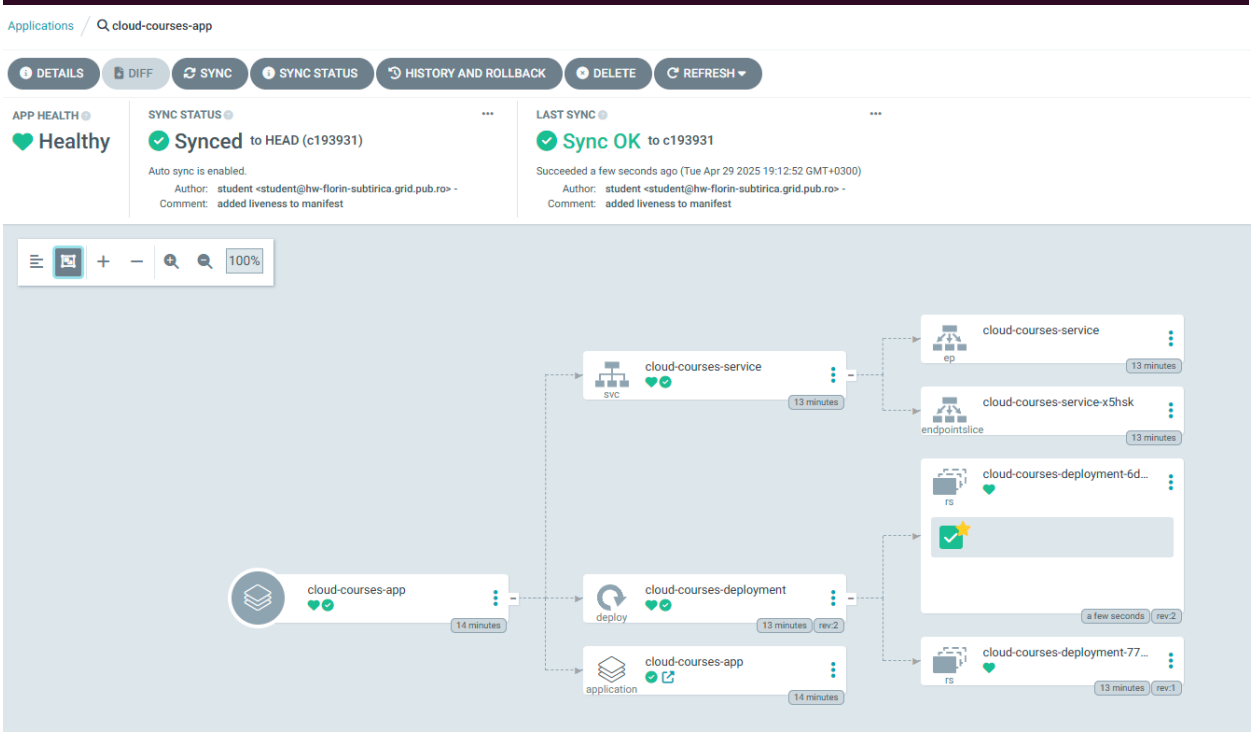
```
student@hw-florin-subtirica:~/nginx-deployment$ vi deployment.yaml
student@hw-florin-subtirica:~/nginx-deployment$ git add .
student@hw-florin-subtirica:~/nginx-deployment$ git commit -m "added liveness to manifest"
[main c193931] added liveness to manifest
 Committer: student <student@hw-florin-subtirica.grid.pub.ro>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 6 insertions(+)
student@hw-florin-subtirica:~/nginx-deployment$ git push
Username for 'http://localhost:3000': florin065
Password for 'http://florin065@localhost:3000':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 411 bytes | 411.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0)
remote: . Processing 1 references
remote: Processed 1 references in total
To http://localhost:3000/florin065/nginx-deployment.git
   059a44e..c193931  main -> main
```



```yaml
    app: cloud-courses
    pod-template-hash: 6d9cdfd76f
  name: cloud-courses-deployment-6d9cdfd76f
  namespace: default
  ownerReferences:
    - apiVersion: apps/v1
      blockOwnerDeletion: true
      controller: true
      kind: Deployment
      name: cloud-courses-deployment
      uid: 31430136-389f-4101-9146-064b5ac31eec
  resourceVersion: '4396'
  uid: 1ca26b5d-c038-4d54-9efe-26c4ed1aadb6
spec:
  replicas: 1
  selector:
    matchLabels:
      app: cloud-courses
      pod-template-hash: 6d9cdfd76f
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: cloud-courses
        pod-template-hash: 6d9cdfd76f
    spec:
      containers:
        - env:
            - name: URL
              value: http://localhost
            - name: BASE_URL
              value: /cloud-courses/
          image: cloud-courses:latest
          imagePullPolicy: IfNotPresent
          livenessProbe:
            failureThreshold: 3
            httpGet:
              path: /cloud-courses/
              port: 3000
              scheme: HTTP
            initialDelaySeconds: 5
            periodSeconds: 10
            successThreshold: 1
            timeoutSeconds: 1
          name: cloud-courses
          ports:
            - containerPort: 3000
              protocol: TCP
          resources: {}
```

```
student@hw-florin-subtirica:~$ kubectl describe deployment cloud-courses-deployment
Name:                   cloud-courses-deployment
Namespace:              default
CreationTimestamp:      Tue, 29 Apr 2025 15:59:25 +0000
Labels:                 app=cloud-courses
                        app.kubernetes.io/instance=cloud-courses-app
Annotations:            deployment.kubernetes.io/revision: 2
Selector:               app=cloud-courses
Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=cloud-courses
  Containers:
   cloud-courses:
    Image:      cloud-courses:latest
    Port:       3000/TCP
    Host Port:  0/TCP
    Liveness:   http-get http://:3000/cloud-courses/ delay=5s timeout=1s period=10s #success=1 #failure=3
    Environment:
      URL:         http://localhost
      BASE_URL:    /cloud-courses/
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  cloud-courses-deployment-7789585c64 (0/0 replicas created)
NewReplicaSet:   cloud-courses-deployment-6d9cdfd76f (1/1 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  15m    deployment-controller  Scaled up replica set cloud-courses-deployment-7789585c64 from 0 to 1
  Normal  ScalingReplicaSet  2m19s  deployment-controller  Scaled up replica set cloud-courses-deployment-6d9cdfd76f from 0 to 1
  Normal  ScalingReplicaSet  2m17s  deployment-controller  Scaled down replica set cloud-courses-deployment-7789585c64 from 1 to 0
```